Collins, Randall. 1979. Chapter 1. *The Credential Society: An Historical Sociology of Education and Stratification*. New York: Academic Press. Permission to reprint granted by the publisher.

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The Myth of Technocracy

There is a naive conception of social history that is extremely popular. People with different viewpoints give it different slants, but the basic story is much the same. The leading character is called Technology, or sometimes Science; very sophisticated storytellers have twin leads called Science and Technology. They are the active agents in the drama. In some versions, they are the heroes; in others, the villains. In all, they are endowed with overwhelming power.

There are some other characters, too. One of them is called Modern Society, who is more or less the dutiful wife, following where Technology leads her. In some accounts she drags her feet; in others she eggs him on. But it does not make very much difference one way or the other because they are married, for better or for worse. There is one other character, a kind of stepchild called the Individual. His job is to fit into the family as best he can. This requires him to be diligent and skillful. Since the family is changing, getting more scientific, technological, and complex all the time, this can be a hard job.

There are a lot of dramatic possibilities here, and our writers have exploited them all.

If it's a success story, one dwells on how the good boy works hard in school and gets his rewards and how the bad boy gets his deserts.[1] This is sometimes attached to a historical yarn about the bad old days before Society met Technology, when the good boys didn't get their rewards because somebody else had already inherited them. Academic storytellers call this the shift from "ascription" to "achievement."

If it's to be a lament, we hear all about how lonely, cruel, and alienating it is to live in this household full of modern hardware.

For do-gooders and alarmists, there is the disturbing tale with the hopeful ending: how some of the stepchildren don't really have a chance to show that they are good boys because they weren't brought up right, didn't have a good school in their neighborhood, or for some reason didn't learn what society required of them. This has been an especially popular story, since it gave people something to rush out and correct when they got through listening to it.

On the other hand, some observers have noticed that opportunities are not becoming more equal, and that the children of the same social classes and racial groups get more or less the same relative rewards as their parents got, regardless of how many efforts people make to give everyone more schooling. This has led to a resurgence of a new "scientific" racism and hereditism, for if the system is just and meritocratic, failure must be the fault of the genes.

For more way-out types, there is the twist in the story in which Technology gets so advanced that it can carry on Society all by itself and the stepchildren don't have to come in from their play after all. This hasn't happened quite yet, of course. Some writers say that all we need is a revolution, others that we just have to wait for our longhaired children to grow up and take command.

For power seekers who feel unduly neglected, there is the pronouncement that Society can no longer be managed by old-fashioned politicians and business executives because it is time for the technical experts to emerge from their laboratories and faculty clubs to take command. Why they haven't done so already is a bit of a mystery, if the story of how things happen in the modern world is otherwise true.

From the point of view of literature, Technocracy isn't a very good story. It has been repeated so many times since it was invented in the eighteenth century by Condorcet and the French *philosophes* that it's gotten to be a bore. Its popularity is obviously owing to something besides entertainment value. One reason might be that the Technocracy story is true. The problem is that it isn't. As we shall see, the sociological evidence is heavily against it, and so are the sociological theories that attempt any sophisticated view of stratifi-

[1]Somehow, the story never talks about the *girl* who works hard in school and gets *her* rewards. May this, although sexist, be a concession to empirical reality?

cation. So it's not truth that upholds the Technocracy story, but rather the appeals of myths.

Yet the Technocracy story does have some facts in its favor, and it is important to see just what they mean. One such fact is that existence of a very considerable amount of technological change over the last two centuries (indeed, even further back) with especially visible effects in the twentieth century on economic productivity and the organization of work. The other fact is the increasing prominence of education in our lives. American sociologists concerned with stratification have concentrated on social mobility, and one main fact has emerged from their research: Education is the most important determinant yet discovered of how far one will go in today's world. Moreover, it has been growing steadily more important in the sense that each new generation of Americans has spent more and more time in school and taken jobs with higher and higher educational requirements. And since schooling has been defined as an agency for meritocratic selection, the rising prominence of education has been the strongest argument for the existence of Technocracy.

Of all those variables that are easily measurable, education (usually indicated simply by the number of years in school) has been found to be the most important predictor of occupational success, along with parents' occupation itself, and much of the effect of parents' occupational level appears to operate through its effect on children's education. The most thorough national survey found a correlation of .60 between occupational level and years of education, as compared to a correlation of .40 between father's occupation and son's occupation, and a correlation of .44 between father's occupation and son's education (Blau and Duncan, 1967:202). This study was able to explain 42% of the variance in occupational attainment; of this, 24% was attributable to education alone, with 18% attributable to father's position, both directly and by its effect on son's educational opportunity. Another methodologically thorough study, a 7-year follow-up of all the 1957 high school graduates in Wisconsin, found education was correlated .62 with job level, whereas father's occupational job level was correlated .33 with son's job level (Sewell *et al.*, 1970). Numerous other studies support this result (Lipset and Bendix, 1959: 189–192; Eckland, 1965; Sewell and Hauser, 1975).

Studies of more specialized occupational groups, such as lawyers (Smigel, 1964: 39, 73–74; Ladinsky, 1967) and scientists (Hargens and Hagstrom, 1967), also show that education is more important than the background influence of father's occupation. Because of such findings, mobility researchers have come to focus primarily on factors determining educational attainment. Much evidence has been amassed to show how class and ethnic advantages and disadvantages operate and to show what are the relative effects of contemporary pressures, such as peer groups, teachers' practices, and the organization of educational bureaucracies (Coleman, 1966; Boudon,

1973; Cicourel et al., 1974; Useem and Miller, 1975). Moreover, these patterns have existed for a long time. The expansion of educational opportunities in twentieth-century America has not much affected the way in which different social classes take advantage of these opportunities; educational attainment has maintained the same degree of correlation with father's occupation and education throughout the last 50 years (B. Duncan, 1967). In view of this failure to overcome class and racial differences, some researchers have raised again the question of innate differences in ability in affecting educational attainment (Block and Dworkin, 1976).

Historical trends show the rising importance of education in people's lives. Table 1.1 shows the rates of educational enrollment for school age groups through the last century. In 1870, 2% of the age group graduated from high school and 1.7% attended college; in 1920, 17% graduated from high school and 9% attended college; by 1970, this had risen to 77% graduating

Table 1.1
Educational Attainment in the United States, 1870–1970[a]

Year	High school students/ population 14–17 years old (%)	High school graduates/ population 17 years old (%)	College students/ population 18–21 years old (%)	B.A.s or first/ professional degree/ population 21 years old (%)	M.A.s or second/ professional degree/ population 25 years old (%)	Ph.D.s/ population 30 years old (%)	Median years of school completed/ population 25 years and older (%)
1870	2.1	2.0	1.7				
1880	2.4	2.5	2.7				
1890	3.6	3.5	3.0				
1900	7.9	6.4	4.0	1.7	.12	.03	
1910	11.4	8.8	5.1	1.9	.13	.02	
1920	26.4	16.8	8.9	2.3	.24	.03	
1930	44.3	29.0	12.4	4.9	.78	.12	
1940	62.4	50.8	15.6	7.0	1.24	.15	8.6
1950	66.0	59.0	29.6	14.8 (1949)	2.43	.27	9.3
1960	87.8	65.1	34.9	14.3	3.25	.42	10.5
1970	92.9	76.5	52.8	21.1	7.83	1.04	12.2

Sources: *Historical Statistics of the United States*, Series A28–29, H223–233, H327–338; *Statistical Abstract of the United States, 1971*, Tables 6, 24, 149, 151, 153, 164, 192, and 205.

[a]All figures involve some degree of estimation from available data, especially for the years before 1900. Note also that these are the ratios of enrollments or degrees awarded in each particular year to the size of the age group in that same year (with the exception of the last column); these are not tabulations of the members of that age group who are enrolled or achieve a degree that year. Hamilton and Wright (1975) present some data that suggest that the college attendance figures are overestimates.

from high school and 53% attending college, with 21% acquiring college degrees. The rising level of educational attainment is typically explained as follows: Industrial society develops through the application of scientific advances to new forms of technology (e.g., Kerr *et al.*, 1960; Clark, 1962; Galbraith, 1967; Bell, 1973). As a result, the school requirements of jobs change. Unskilled positions are greatly reduced, first with the decline of agriculture, then of heavy manual labor; skilled positions become proportionately more common with the rising demand for skilled technicians, clerical workers, and professional specialists. The requirements for administrative leadership positions are upgraded too, in the larger, more complex, and technically more innovative modern organizations. According to this explanation, elites must come to depend on skill rather than on family background or political connections. In general, modern societies move away from ascription to achievement, from a system of privilege to a technical meritocracy.

One implication of this analysis is that educational requirements for jobs change over time. The point is not often documented with systematic evidence, however, and hence certain ambiguities remain as to just what changes in "requirements" mean. As we shall see, the evidence as to what education is *actually necessary* for jobs does not necessarily support this thesis. However, it is possible to show what requirements *employers* have set for jobs at different periods (see Table 1.2). A somewhat diverse set of samples makes precise numerical estimates unreliable, but the general trends are clear. In 1937–1938, 11% of employers required high school diplomas or more for skilled laborers; in the 1945–1946 period, this had risen to 19%; and in 1967 to 32%. For clerical jobs, the percentage requiring high school or more rose more slowly: 57% in 1937–1938; 56–72% in 1945–1946; 72% in 1967. For managers there is a dramatic postwar rise, from 12% of employers requiring college degrees in 1937–1938 (68% in the 1945–1946 sample), to 44% in 1967.

For earlier periods we must use estimates. It seems that formal schooling was of little importance through the mid-nineteenth century. This was true even for the professions, where apprenticeship training prevailed. Toward the latter part of the century, college training became necessary for the major professions; high school degrees became common among major businessmen and among minor professionals such as teachers. After World War I, professions had established firm requirements of higher education; college education in any field became the norm among major businessmen, and high school degrees began to become the criterion for clerical work. Secondary and elementary teaching now required a college degree. The Depression saw a great expansion of the school system, and after World War II these standards became less preferential and more rigorously required. Postcollege training became required by the traditional elite professions and began to

Table 1.2

Percentage of Employers Requiring Various Minimum Educational Levels of Employees, by Occupational Level, 1937–1938, 1945–1946, and 1967

Education required	Unskilled	Semiskilled	Skilled	Clerical	Managerial	Professional
National survey, 1937–1938[a]						
Less than high school	99	97	89	33	32	9
High school diploma	1	3	11	63	54	16
Some college				1	2	23
College degree				3	12	52
Total (%)	100	100	100	100	100	100
New Haven, 1945–1946[b]						
Less than high school	96	96		81	44	43
High school diploma	4	4		17	50	40
Vocational training beyond high school				2	4	1
Some college					2	10
College degree						6
Total (%)	100	100		100	100	100
Number surveyed	62	74		78	123	107
Charlotte, N. C., 1945–1946[c]						
Less than high school	100	86		81	28	30
High school diploma		14		18	65	50
Vocational training beyond high school					4	3
Some college				1	3	9
College degree						8
Total (%)	100	100		100	100	100
Number surveyed	67	91		99	104	104
San Francisco Bay area, 1967[d]						
Less than high school	83	76	62	29	27	10
High school diploma	16	24	28	68	14	4
Vocational training beyond high school	1	1	10	2	2	4
Some college				2	12	7
College degree					41	70
Graduate degree					3	5
Total (%)	100	101	100	101	99	100
Number surveyed	244	237	245	306	288	240

[a]Source: Bell (1940: 264), as analyzed in Thomas (1956: 346). Bell does not report the number of employers in the sample but it was apparently large. Totals differing from 100% are due to rounding.

[b]Source: Computed from Noland and Bakke (1949: 194–195).

[c]*Ibid.*

[d]Source: Survey conducted by the Institute of Industrial Relations, University of California at Berkeley, 1967.

become common among engineers and business managers. The college degree became required for managers in major business and government organizations. Clerical work, becoming confined largely to females, still remained around the norm of a high school degree, although the educational level of such workers began to move beyond this. High school graduation, and sometimes vocational training after high school, began to become required for manual positions, most commonly at skilled levels but occasionally at lower job levels as well. For example, high school degrees were required by 17% of employers of unskilled laborers in 1967, and the figure is probably higher today.

At the same time, educational requirements have become more specialized. Business administration training for managers was virtually unknown in 1900 and put in only a scattered appearance in the 1920s. With the advance of college enrollments and employers' educational requirements after World War II, however, business administration has taken on a new importance; in a 1967 survey done by the Institute for Industrial Relations at the University of California, Berkeley (see Table 1.2) researchers found that 38% of those employers who required college degrees preferred business administration students and another 15% preferred engineers. More recent data, if it were available, would probably show even more emphasis on specialization.

The explanation for these trends has commonly been treated as obvious: Education prepares students in the skills necessary for work, and skills are the main determinant of occupational success. That is, the hierarchy of educational attainment is assumed to be a hierarchy of skills, and the hierarchy of jobs is assumed to be another such skill hierarchy. Hence education determines success, and all the more so as the modern economy allegedly shifts toward an increasing predominance of highly skilled positions.

In this chapter and in the one that follows, I will break down this theory into its component parts and examine them against the relevant empirical evidence. This has rarely been done, although the evidence on most points is plentiful enough; it has only to be drawn together from disparate and perhaps overspecialized academic studies. When one does so, the technocratic model crumbles at virtually every point where one can test it against empirical evidence in a more detailed fashion. The educationocracy, which is its backbone, is mostly bureaucratic hot air rather than a producer of real technical skills. Whichever way we look at it—comparing the performance of more educated and less educated people at work, finding out where vocational skills are actually learned, examining what students pick up in the classroom and how long they remember it, examining the relationship between grades and success—the technocratic interpretation of education hardly receives any

support. The same is true if we test the functional theory of stratification, of which the technological theory is a particular application.

We are left in the position of looking for an explanatory theory that will work. How do we account for the fact that modern America has come to be stratified around an educational credential system with a stranglehold on occupational opportunities and a technocratic ideology that cannot stand close examination?

In the last few years, there has been an undercurrent of radical questioning of the dominant technocratic model. But those who raise this criticism have not displaced the technocratic model because they do not really disprove it. They present plausible alternative interpretations of some of the same facts upon which technocratic theory bases itself, and in some cases add useful new evidence and new theoretical schemes. These radical critiques are not conclusive for another reason as well: They do not agree among themselves upon the particular mechanisms of an alternative explanation of educational stratification.

The most limited critique is that of Jencks *et al.* (1972). Based upon the analysis of aggregate census and survey data, they show that a good deal of career stratification is produced other than by education (some 60% of the variance is left unexplained and might just as well be attributed to chance, they argue). Examining historical trends and hypothetical projections, they claim that changes in the distribution of education do not produce changes in the distribution of rewards, and hence that educational reform is not the answer to reducing economic inequality but a diversion from that task.

The main results seem to be accurate. But the analysis leaves some central questions unanswered: Why is it that *some* stratification is correlated with, and presumably produced by, educational attainment? And why is it that this overall stratification structure exists in which increasing numbers of people have sought increasing amounts of education, even if their relative stratification positions on the whole have not changed? These are not trivial questions, for if one can understand how education contributes to stratification, even if it explains only part of the pattern, it may give us the paradigm for a more general model that explains most of stratification rather than leaving most career outcomes to an unexplicated concept of "chance."

Illich (1970), on the other hand, claims that education is indeed central to economic stratification. Education, he asserts, is not the basis of technical skills, but it does serve as a means by which opportunities to practice particular forms of work are monopolized, and hence a basis for restricting access to the actual on-job acquisition of skills. For him, the highly credentialized professions are the epitome of modern stratification. His solution for creating greater equality, then, is deschooling: to eliminate formal, compulsory education as it now stands and substitute opportunities for on-job experience in any

desired occupation. The argument that education is an artificial device for monopolizing access to lucrative occupations is plausible, and it may well be that deschooling would produce an actual change in economic inequality. In both respects, the theory makes stronger claims than Jencks *et al.* But it leaves open the explanatory question: If education is technically irrelevant, by what means has it proven possible for it to monopolize job opportunities? Why do employers accept it? What general mechanism is involved?

The most systematic explanations have come from Marxian-oriented sociologists and economists. Unfortunately, these tend toward two opposing versions of the mechanism of educational stratification.[2]

On one side, there is the position found among French thinkers who argue that education serves to reproduce the class relations of capitalism. Althusser (1971) gives a very abstract argument that education serves to reproduce the social relations of production (as opposed to the material means of production), especially by providing ideological legitimation for class domination.[3] Bourdieu and his colleagues (Bourdieu and Passeron, 1964, 1977; Bourdieu *et al.*, 1974) give an empirical version of the class reproduction argument. By presenting data on the educational careers of children of different social classes, they are able to describe how class advantages are passed along through the meritocratic educational system itself. The key concept is "cultural capital," a set of cultural outlooks and predispositions that children receive from their home environment and invest in formal education. This capital determines their progress through the schools and is cumulatively enhanced or diminished, according to the previous accumulation of cultural capital, at any transition point among different levels and types of schooling within the system. Thus the older system of direct inheritance of material property has been supplanted by a system of indirect material inheritance through the direct inheritance and investment of cultural property.

The specifics of Bourdieu's reproduction argument are quite plausible. But the model does not actually disprove a technocratic interpretation of education-based work skills or even a biological-hereditist explanation of the transmissions of class advantages. In fact, Bourdieu *et al.* present the same kind of data from which American sociologists have traditionally argued

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[2]Not all Marxian and left-oriented theories have rejected the technocratic model, however. O'Connor (1973) accepts the technocratic interpretation of education; the skills education allegedly provides are one of the infrastructural supports that the modern state provides for capitalists, and hence one of the costs contributing to the fiscal crisis. Giddens' (1973) eclectic approach includes education-based work skills as one basis of stratification.

[3]Similarly, Habermas (1970), Althusser's German counterpart in Marxist philosophical sociology, asserts that science and technology are *ideologies* by which modern stratification is legitimated and the real bases of class interest are hidden. The argument is not an empirical one, but is based on analysis of the ideas of science and technology in themselves.

technocratic claims or mildly reformist interpretations of the need to expand educational access, as well as racist-hereditist interpretations. Direct evidence to disprove the empirical contentions of the technocratic theory is needed to show that an alternative explanation is superior. Also, the larger mechanism explaining the macro pattern of educational stratification and its historical development is obscure in Bourdieu's model. To claim that this form of self-perpetuating "cultural capitalism" comes into being to protect modern industrial capitalism is formally similar to the functional model of a self-equilibrating system; only the moral tone changes from adulation of the system, in the case of American functionalists, to condemnation of it, in the case of the French. To provide an explanation of why structures develop as they have historically, one needs to compare divergent structural cases, and these raise a question as to whether capitalism per se is a sufficient explanation—as an even more exclusive reliance on stratification by culture (education) can be found in most of the communist countries today.

Another alternative that is not ruled out by the cultural capital model is contained in the claims made on behalf of black, Chicano, and other minorities that education is a form of cultural imperialism (Valentine, 1971; Carnoy, 1974). Minority members are kept in inferior class positions because they compete in a school system in terms of a culture that is alien to their own, and hence they compete at a disadvantage. The argument is most plausible for the United States and seems hardly applicable to the more ethnically homogeneous countries of Europe and elsewhere. But it cannot be ruled out as at least a partial explanation of one possible case, as the United States has a much more highly developed educational system than any other country in the world. The argument does illustrate the general difficulty of definitively accepting or rejecting plausible explanations based on showing the *results* of education for stratifying people, rather than by giving a causal explanation of how a stratification system works under particular conditions to produce various organizational structures and distributional outcomes.

The most novel and best documented work in an explicitly Marxian vein is that of Bowles and Gintis (1976). But their results and interpretations give a mechanism of educational stratification the obverse of that presented by Bourdieu's cultural capital argument (and implicitly by the ethnic minority critics as well). Bowles and Gintis present evidence that education produces compliant, disciplined workers; historically, it has been imposed on workers from above. But this contradicts the conception of Bourdieu that education is a *possession*, valued above all by the higher classes for their own benefit.[4] If

[4]Bernstein's (1971, 1973, 1975) interpretation of education tends to bolster, and give a specific interpretation of, Bourdieu's concept of education as cultural capital. Bernstein sees education as selecting and rewarding the cognitive and linguistic traits of the white-collar class, as

education is an imposition of control upon others, why is it that people compete to attain education, thereby expanding the educational system in the United States far beyond any other in the world? Why is it that the upper class and the independent, professional, upper middle class have the most education, while the order-taking working class has the least? Yet Bowles and Gintis's evidence of the psychological effects of education are strong. The contradiction is not on the level of their data but on their interpretation of it.

In sum, there are two difficulties with the critical theories of education:

1. If education leads to stratification, how does it do this? Is it a weapon or resource in the hands of those who possess it, and if so, of what nature? Is it technical skills (which most such theorists reject) or cultural capital? If the latter, how does this operate? Is it Bernstein's linguistic skills (themselves very close to functional work skills), or sheer prestige (ethnic or otherwise), or an arbitrary bureaucratic credential? Or, on the contrary, is education not a possession but an imposition, a mark of socialization into subservience?

2. These are questions of process *within* a system of educational stratification. The question also arises, Why does the *structure* take that particular form—why do educational requirements arise at particular times in history, and why do the structures of the educational systems and of the organization of occupational careers vary as they do among different countries of the modern (and premodern) world?

In what follows, I shall systematically review the evidence regarding the place of education in the economy generally and in individual careers specifically. The evidence comes from a variety of approaches. In this chapter, I review aggregate data on shifting educational levels, job skill levels, and economic development; I also review here correlations of education with on-job productivity, the sources of vocational training, and studies of what is actually learned in school. Chapter 2 will deal with studies of organizations, where work actually takes place. This will show what we know of technology's effects on work, of the struggles for control and advancement within organizations, and how education fits into these processes.

In all this, two themes become steadily more prominent: (*a*) the place of education as a cultural basis of group formation, especially for groups strug-

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against the different cognitive–linguistic traits of the working class. Bernstein's interpretation, though, is not far from a technocratic one; these cognitive–linguistic traits are regarded as work-relevant skills based on the demands of jobs in industrialism (and increasingly, in late industrialism); the only difference from the usual meritocratic interpretation is that Bernstein sees these as directly transmitted via the family and only reinforced in the school system. Bourdieu's concept of cultural capital, on the other hand, seems to represent more of the qualities of a leisure "high culture."

gling to shape their occupational positions and careers; and (*b*) the role of technology in setting the problems and material rewards around which these struggles hinge. In Chapter 3, I will outline a general theory of the interrelations between the two realms of material production and cultural domination, within which educational stratification finds its place. The second half of the book then goes on to apply this perspective to the historical development of educational stratification in the United States. A more complete analysis would be more comparative and would show the conditions that have produced different sorts of educational structures in various parts of the world.[5] But even with a more limited historical focus, it should be possible to see how successful this approach is in accounting for the macro-structural patterns as well as the micro processes of educational stratification in America.

THE TECHNOLOGICAL RELEVANCE OF EDUCATION

The technological function theory of education may be stated in terms of the following propositions:

1. The school requirements of jobs in industrial society constantly increase because of technological change. Two processes are involved:
   a. The proportion of jobs requiring low skill decreases and the proportion requiring high skill increases.
   b. The same jobs are upgraded in skill requirements.
2. Formal education provides the training, either in specific skills or in general capacities, necessary for the more highly skilled jobs.
3. Therefore, educational requirements for employment constantly rise and increasingly larger proportions of the populace are required to spend longer and longer periods in school.

Proposition 1a. *Educational requirements of jobs in industrial society increase because the proportion of jobs requiring low skill decreases and the proportion requiring high skill increases.*

Available evidence suggests that this process accounts for only a minor part of educational upgrading, at least in a society that has passed the point of initial industrialization. Folger and Nam (1964) found that 15% of the increase in education of the U.S. labor force during the twentieth century may be attributed to shifts in the occupational structure: a decrease in the proportion of jobs with low skill requirements (unskilled laboring and service positions)

[5] I have outlined some part of this elsewhere, especially for historical cases prior to the twentieth century, in Collins (1977).

and an increase in jobs with high skill requirements (skilled manual, professional, technical, and managerial positions). The bulk of the educational upgrading (85%) has occurred within job categories.

Proposition 1b. *Educational requirements of jobs in industrial society rise because the same jobs are upgraded in skill requirements.*

The only available evidence on this point is Berg's study of data collected by the U.S. Department of Labor in 1950 and 1960, which judges the amount of change in skill requirements of specific jobs (1970: 38–60). Berg compares these shifts in skill requirements with overall changes in educational levels in the labor force. For several reasons, the results are not entirely clear-cut. Berg's analysis does not separate changes in the proportions of different kinds of jobs from changes in skill requirements of the same jobs. If we use Folger and Nam's figure of 15% as an estimate of changes in *kinds of jobs,* then most of Berg's evidence on skill levels shift must refer to changes *within* jobs, although the overall shifts in his data attributable to such internal job changes must be curtailed somewhat. Another ambiguous point is how much skill particular levels of education must be regarded as providing. Berg gives a number of alternative models embodying different assumptions. Under the most plausible ones, it appears that the educational level of the labor force changed somewhat in excess of that necessary to keep up with the skill requirements of jobs between 1950 and 1960. Overeducation for available jobs was found particularly among males who had graduated from college and females with high school degrees or some college, and overeducation appears to have increased during the decade 1950–1960. Whether this decade was a typical one, of course, remains to be seen. In any case, just how much the education "necessary" for employment has changed due to changes in skill levels as against other causes cannot yet be settled by quantitative measures until we know just how much skill is embodied in education and whether skills can be acquired in any other way. Evidence on this point will be taken up below.

Proposition 2: *Formal education provides required job skills.*

This proposition may be tested in two ways: (*a*) Are better educated employees more productive than less educated employees?; and (*b*) Are vocational skills learned in school or elsewhere?

Are Better Educated Employees More Productive?

The evidence most often cited for the productive effects of education is indirect, consisting of relationships between *aggregate* levels of education within a society and its overall economic activity. These are of three sorts.

1. The national growth approach involves calculating the proportion of the growth in the U.S. gross national product attributable to conventional inputs of capital and labor (Schultz, 1961: 1–16; Denison, 1965: 328–340). These inputs leave a large residual that is attributed to improvements in skill of the labor force based on increased education. This approach suffers from the difficulty in distinguishing among technological change affecting productive arrangements, changes in the abilities of workers acquired by the experience of working with new technologies, changes in skills due to formal education, and motivational factors associated with a competitive or achievement-oriented society. To assign a large proportion of the residual category to education is basically arbitrary. Denison makes this attribution because persons with higher levels of education have proportionately higher incomes, which he interprets as rewards for their contributions to productivity. Yet, although it is a common assumption in economic argument that wage returns reflect output value, wage returns cannot be used to prove the productive contribution of education without circular reasoning.

2. International correlations of levels of education and level of economic development show that the higher the level of economic development of a country, the higher the proportions of its populations in elementary, secondary, and higher education. Harbison and Myers (1964) divided countries into four groups:

> Level I comprises 17 countries (almost all in Africa) with an average per capita GNP of $84.
> Level II comprises 21 countries (all in Asia and Latin America) with average per capita income of $182.
> Level III includes 21 countries (mostly in southern and eastern Europe, plus the richest Asian and Latin American countries) with an average per capita GNP of $380.
> Level IV includes 16 countries (Europe plus the English-speaking countries) with average per capita GNP of $1,100.

These four groups correlate .89 with a composite index of enrollment in schools of all grades.

But *within* each of these four economic levels, there is no significant correlation between economic development and education. Within Level IV, for example, Denmark and Sweden have the lowest education enrollment indices (77 and 79 respectively, as compared to a median of 105 for the whole group and a high of 261 for the United States) but have per capita GNPs well above the median; conversely, France, Japan, and the Netherlands have educational indices above the median (108, 111, and 134) but per capita GNPs below the median. Many of these variations are explicable in terms of political demands for access to education (Kotschnig, 1937; Ben-

David, 1963–1964: 247–330; Hoselitz, 1965: 541–565). Even the overall correlations of educational enrollment with levels of economic development do not settle the question of causality: Which causes which? Education may be a luxury that richer countries can afford or that more democratic governments are made to provide by the people. The overproduction of educational personnel in countries whose level of economic development cannot absorb them suggests that the demand for education need not come directly from the economy and may run counter to economic needs.

3. Time lags of education and economic development can help settle the question of the direction of causality. Such correlations show that increases in the proportion of the population in elementary school precede increases in economic development (Peaslee, 1969: 293–318). The take-off point occurs once 30–50% of the 7- to 14-year-old group is in school. Peaslee suggests similar anticipations of economic development for increases in secondary and higher educational enrollment, but the data do not support this conclusion. The pattern of advances in secondary school enrollments preceding advances in economic development is found in only a small number of cases (12 of 37 examined by Peaslee). The pattern in which growth in university enrollments is followed by subsequent economic development is found in 21 of 37 cases, but the exceptions (including the United States, France, Sweden, Russia, and Japan) are of such importance as to cause serious doubt on any *necessary* contribution of higher education to economic development. The main contribution of education to economic productivity, then, appears to occur at the level of the transition to mass literacy and not significantly beyond this level.

Direct evidence of the contribution of education to *individual* productivity is summarized by Berg (1970: 85–104, 143–176). It indicates that better educated employees are not generally more productive and in some cases are less productive than others among samples of American workers at various levels. The levels of education being compared, of course, are relative to a particular range. Among factory workers (both male and female), maintenance men, and department store clerks, comparisons are between various levels of education at or below the high school degree. Among bank tellers, secretaries, factory technicians, insurance salesmen, airport controllers, and military technical personnel, comparisons centered on high school versus some college training or college degrees; in these cases, where differences did exist, workers with high school training were found to be more productive than persons with more education, but they were also more productive than persons with less education. In a number of these samples, education was not related to performance at all. Berg also includes a study of engineers and research scientists in major electrical manufacturing industries; this contrasted educational qualifications from below the B.A. level through the Ph.D., and

found the average performance evaluation of Ph.D.s to be the highest, but no differences among below-B.A., B.A., and M.A. levels.

This evidence does not show that persons with no schooling at all are better employees than those with college or advanced degrees; since virtually no such totally uneducated employees are available today, the comparisons are simply not made. Most plausibly, given the necessity for literacy and certain cognitive skills in these jobs, certain minimal educational qualifications do exist. What Berg's data do show is that job skill cannot be assumed to be related to educational levels in a simple linear fashion, and that beyond certain plateaus the relationships may well be reversed. Berg also shows that better educated employees are more likely to be dissatisfied with jobs and to change jobs more often than less educated persons—evidence that education may be primarily regarded as a matter of status by those who possess it.

Are Vocational Skills Learned in School or Elsewhere?

There are a number of ways to approach this question. For some specific manual skills, there are studies of on-the-job and vocational school training. There is also evidence of how retraining is carried out in organizations experiencing technological changes. For higher level jobs, evidence is less direct. As we shall see, it is plausible that many of the skills for the learned professions, as well as for other high-level posts, are acquired on the job, but the existence of legal requirements for degrees (e.g., in medicine, law, or pharmacy) makes a comparison group of noneducated practitioners unavailable, at least in the modern era. As a substitute, we may examine evidence on what is learned in schools and what bearing school grades have on subsequent occupational performance.

Vocational Education

Vocational education in the schools for manual positions is virtually irrelevant to job fate. A 1963 study showed that most skilled workers in the United States acquired their skills on the job or casually (Clark and Sloan, 1966: 73). Of those (41% of the total) who did have some formal training, over half (52%) received this in the armed forces, in apprenticeship programs, or in company training rather than in public schools, commercial schools, or correspondence courses. Persons who do have formal vocational education do not generally fare better than those without it; graduates of vocational education programs are no less likely to be unemployed than high school dropouts (Plunkett, 1960: 22–27; Duncan, 1964: 121–134). A major reason for the failure of vocational schooling is probably that vocational high schools are

known as places where youthful troublemakers are sent to remove them from the regular schools. The warfare between teachers and students at a regular high school is considered mild compared with the real gang-type violence (often with ethnic or racial overtones) that is reputed to occur at "Tech." Even if a vocational student happens to learn some usable skills, his or her attendance at a vocational high school is likely to be taken by the discerning employer as a sign of bad character.

Retraining

A 1967 survey of 309 employing organizations in the San Francisco Bay area provides information on retraining in response to technological change.[6] Retraining for technological change appears to be surprisingly easy. The vast majority of organizations (84% in this survey) were able to retrain their own labor force informally on the job in a period of not more than 3 months. Formal education was used little in retraining, and when it was used, 90% of the organizations completed it in less than 3 months. In one very modern and highly innovative chemical plant, a major technical change—the building of a new type of power plant—required only that one man be sent out for 2 months of technical training. This plant seemed to be on a technological plateau. It already had relatively high educational requirements and had been undergoing technological changes constantly for many years. As a result, changes were handled internally without changes in educational requirements for employment; the only trend reported by the industrial relations manager was the promotion of semiskilled workers to maintenance jobs. Such technological plateaus may have been reached by many (or most) organizations in this sample. Previous studies have shown similar patterns: relatively easy retraining for technological change, although sometimes accompanied by decreases in employment for less skilled workers (Bright, 1958: 85–97).

What Is Learned in School?

There has been little study of what is actually learned in school and how long it is retained. What evidence is available, though, suggests that schools are very inefficient places of learning. Many of the skills used in managerial and professional positions are learned on the job, and the lengthy courses of study required by business and professional schools exist in good part to raise the status of the profession and to form the barrier of socialization between practitioners and laymen. The degree to which education provides mainly

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[6]Organizations employing 100 or more persons were sampled from all industry groups. Further information on this survey is provided in Collins (1969) and Gordon and Thal-Larsen (1969).

these status and socialization functions varies among occupations. It seems to be great in pharmacy, where the 4-year university course is felt by the practitioners to be "good for the profession," but in which a few months' training is sufficient for work competence (Weinstein, 1943: 89). Schools of education are often criticized as requiring courses of little value to the actual practice of teaching. Schools of medicine and law include many requirements that practitioners admit are forgotten in later practice and do not include many of the essentials that are learned through experience and coaching on the job. Business schools are increasingly used as a recruitment channel for managers, but their level of instruction is not generally high. Most business schools prepare their students for a particular first job rather than teaching skills that may be useful throughout the career. Moreover, the particular kind of training taken often turns out to be irrelevant, as most graduates find jobs outside of their own specialty (Pierson, 1959: 9, 55–85, 140; Gordon and Howell, 1959: 1–18, 40, 88, 324–337).

There are also indications that most students, at least in the United States, do not learn very much in school. Learned and Wood, testing college students at 2-year intervals in the 1930s, found very small increments in their performance on standardized tests in their areas of study, suggesting that very little of what is learned in particular courses remains even through the next few years (1938: 28). More recent applications of achievement tests to high school students found that average levels of proficiency in reading, science, and mathematics are far below what is considered the norm for such levels. International comparisons of mathematics proficiency scores of high school seniors found American students the lowest of 12 countries, with an average score of 13, as compared to scores of 22 to 37 for the other 11 countries.[7]

These results should be seen in the context of studies of what happens in American schools. From the 1920s through the 1960s, observational and survey research has shown that the primary focus of attention in schools has been on nonacademic concerns (Waller, 1932; Coleman, 1961; Holt, 1964; Becker *et al.*, 1968).[8] Teachers have been mainly concerned with maintaining authority in an ongoing struggle with the students. Athletics and other extracurricular activities are stressed by teachers and administrators as well as by students, especially for their ritual functions in maintaining school order. At college as well as at secondary levels, close studies of what actually happens find students preoccupied with strategies for achieving grades with a

[7]*National Assessment of Educational Progress* (1970); Husen *et al.* (1967). It is claimed that the top American students are comparable to those elsewhere. But this is beside the point. Clearly the massive American educational system does not provide *widespread* scientific skills "demanded" by a high stage of technological development exemplified in America.

[8]See Bidwell (1965: 972–1022) for a summary of organizational studies.

minimum of learning. As in other formal organizations, there is a struggle in schools over standards of evaulation and much goal displacement away from official aims in the direction of meeting the informal exigencies of keeping the institution running.

The content of public school education has consisted especially of middle-class culture rather than academic skills per se. The nineteenth-century schools emphasized more than anything else strict obedience to old-fashioned standards of religious propriety. The Progressive reforms resulting from the influx of working-class children through compulsory attendance laws substituted milder forms of socialization, such as student government and supervised activities. Studies from the 1920s through the 1960s have indicated the general success of such programs in involving at least the middle-class students, plus an upwardly mobile minority of others, into these adult-controlled imitations of middle-class adult sociability (Waller, 1932: 15–131; Becker, 1961: 96). Their success is also indicated by the widespread inculcation of nonpartisan but highly idealized beliefs about the political system among American school children (De Charms and Moeller, 1962: 136–142; Hess and Torney, 1967). In sum, what is learned in school has much more to do with conventional standards of sociability and propriety than with instrumental and cognitive skills.

Grades and Success

Such observations are relevant to interpreting the data on the relationships between grades and subsequent occupational performance. Evidence on this point is mixed. A 20-year follow-up of high school graduates in several cities found that grades in high school were not related to subsequent income, unless the student went on to graduate from college (Wolfle and Smith, 1956: 201–232). A study of living college graduates in 1947–1948 found that men who reported mostly A grades in college had higher incomes than others, but that the differences among B, C, and D students were negligible (Havemann and West, 1952, Charts 37–38). Among women, however, there were few differences by grades even at the A level, except that women in nonteaching professions who had A records earned somewhat more than others. A study of Dartmouth College alumni from the class of 1926, followed up 30 years later, used college records rather than personal recollections of grades; it also reported that students with the highest grades reported the highest incomes (Husband, 1957: 157–158). This result was confined to a contrast between the very best students (those with grade point averages of 3.10 and over on a scale of 0–4.00) and all the others; among the remaining (with grade point averages from 1.7 to 3.09), there were no differences in income. Even sharper differences in income were found between those with high and low

participation in extracurricular activities, especially in campus politics; intercollegiate athletics were also important for later success. A 1951 study of Fresno State College graduates, on the other hand, found no relationships between grades and income (Jepsen, 1951: 616–628). Similarly, a 1963 follow-up of a national sample of a 1958 college graduating class found little relationship between grades and subsequent occupational success (Sharp, 1970: 110).

Other studies show very low correlations between college grades and success for students trained in business, engineering, medicine, school teaching, and scientific research (Goslin, 1966: 153–167; Jencks and Riesman, 1968: 205). Medical school grades bear no relationships to success in various kinds of medical practice (Price *et al.*, 1963). For officers graduating from military academies, academic record makes little or no difference to subsequent evaluation or promotion; instead, participation in intercollegiate athletics has been the best predictor of success (Janowitz, 1960: 134–135). Among business students as well, participation in extracurricular activities has been generally more important than grades (Gordon and Howell, 1959: 79–80). A 5-year follow-up of 1963 business school graduates found no correlations between occupational success and either grades, instructors' ratings, or school achievement tests (Cox, 1968). For engineers, high college grades and degree levels generally predict high levels of technical responsibility and high participation in professional activities, but not high salary or supervisory responsibility (Perrucci and Perrucci, 1970: 451–463).

The most reliable finding is that grades are good predictors of subsequent *academic* performance. Thus high school grades predict college grades, and college grades predict admission to graduate school (Holland and Nichols, 1964: 55–65; Wegner, 1969: 154–169). It is probably for this reason that grades bear some relation to occupational success, especially in the Haveman and West 1948 study of college students (1952) and among the Dartmouth students (Husband, 1957); grades are particularly important in providing access to the advanced training required for the high-paying professions. Thus a follow-up study of students entering the University of Illinois in 1952 found that high school grade rank had a small effect on occupational attainment, mainly because the better students went on to professional training (Eckland, 1965).

It appears, then, that grades are tied to occupational success primarily because of the certification value of educational degrees rather than for the skill (often negligible) that they themselves may indicate. It is known that employers generally do not select employees on the basis of their school grades; rather, they look for the completion of a degree in particular subjects and especially for diffuse "personality" characteristics (Thomas, 1956: 356–357; Drake *et al.*, 1972: 47–51). Grades, and the capacity to get them, operate as specialized forms of control within the school system itself, reflect-

ing the teacher's judgment on students' compliance with instruction. The sheer amount of school work done seems to be the best predictor of high grades, with the highest grades reserved for work done in amounts beyond what is asked for (Sexton, 1961: 279–280). It is also known that grades are not related to measures of creativity and possibly bear some inverse relation to creativity. Teachers prefer and reward the hardworking, compliant students and tend to dislike the more innovative ones, especially if they do not respect routine discipline (Getzels and Jackson, 1962; Torrance, 1964).[9]

It is also known that grades are highly correlated to social class background (Sexton, 1961: 25–86). Strikingly enough, when class background factors are held constant by statistical analysis, differences in achievement scores among schools by quality of teaching, facilities, or by the amount of funds spent on them disappear (Coleman, 1966: 290–330). This suggests that schools have relatively little effect on learning, except insofar as they mold those disciplined cultural styles already prominent among the higher social classes; grades simply reward and certify displays of middle-class self-discipline.

In sum, shifts in the proportions of more skilled and less skilled jobs do not account for the observed increase in the educational level of the American labor force. Economic evidence indicates no clear contributions of education to economic development beyond the provision of mass literacy. Education is often irrelevant to on-the-job productivity, and is sometimes counterproductive. Specifically vocational training seems to be derived primarily from work experience rather than from formal school training. The actual performance of schools themselves, the nature of the grading system and its lack of relationship to occupational success, and the dominant ethos among students suggest that schooling is very inefficient as a means of training for work skills.

The evidence so far has focused on education. But it is also possible to investigate the nature of work itself and the organizational relationships within which it takes place. This is the subject of Chapter 2.

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[9]Higher creativity scores are found among students in conflict-ridden urban schools than in conformist suburban and small-town schools. See Boyle (1969: 71–90).